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EXAM DOCUMENTATION

Dear Sigmaritan,

Hope you enjoyed the learning and exploration of the Artificial Intelligence field, and you found this field as attractive as we all do. But now it's time to put all the knowledge you've accumulated into practice. This examination was created with the aim of testing your knowledge and finding the level of understanding of the Machine Learning and Data Analysis principles.

The exam will consist of two parts:

1. The exploration of the dataset and the training of a Machine Learning model on it.
2. The presentation of the insights and results obtained from the work you have done on this data set in a PowerPoint presentation in front of the jury.

The first part should be submitted by loading all the notebooks into a GitHub repository and sending the link to that repository to Elena Graur by the 28th of April 2026, 23:59.

The second part will be presented individually by each Sigmaritan, that is examined by the jury, consisting of the members of the Castor and Pollux teams. The date of the presentation will be agreed upon personally with each person and the jury.

More details about the organisational part of the exam can be found on this Pandora page.

TASK#1.

In the archive, you have a CSV file. This file is the data that you should use to pass the exam.
The client who needs your help provided a dataset which contains a comprehensive University football injury record, offering detailed insights into player health risks and injury patterns in university athletics. It provides useful information about the player characteristics, training loads, physical metrics, performance indicators, and injury history to facilitate understanding of injury causation and prevention in university-level football programs. Your task, as follows, is to try to predict whether a player is at risk of having an injury the next season. The dataset has the following fields:

- Age;
- Height_cm;
- Weight_kg;
- Position;
- Training_Hours_Per_Week;
- Matches_Played_Past_Season;
- Previous_Injury_Count;
- Knee_Strength_Score;
- Hamstring_Flexibility;
- Reaction_Time_ms;
- Balance_Test_Score;
- Sprint_Speed_10m_s;
- Agility_Score;
- Sleep_Hours_Per_Night;

- Stress_Level_Score;
- Nutrition_Quality_Score;
- Warmup_Routine_Adherence;
- Injury_Next_Season;
- BMI.

In this task, you should:

1. Load and analyse the dataset, and try to extract insights from the data.
2. Prepare the data for a machine learning algorithm-ready format.
3. Handle the NaN values in the dataset.
4. Apply some post-processing and post-analysis techniques, such as Feature Selection, Feature Engineering, or others and see if they can help you.
5. Create a model or a Pipeline that will show the best performance in predicting the target column.

The target column is **Injury_Next_Season**. Try to create models that will predict this value.

You MUST show each step of your implementation, even if you consider it "not so important". The goal of this examination is to see what you were able to learn during your study process. The process itself matters more than the final result!

Some more advice:

- You can split different steps of your work into separate notebooks.
- Commend the code and use the markdown cell type to explain what or why you are doing.
- Explain in written form the choices and decisions that you made during the processing of the data.
- Use plots, but don't abuse them.

- Don't chase a perfect model; better show us why you chose a specific model as the best one.

TASK#2.

Your second task is to create and prepare a presentation on the work that you have done. In the presentation that you should explain the work that you have done and why you made the choices that you made during the analysis and Machine Learning model development. Some more pieces of advice for the presentation are:

- KISS (Keep It Simple Stupid) - keep the presentation as simple as possible.
- Give some context about the field on which your data set is about.
- Try to avoid as much as possible having a lot of text on your slides.
- Try to tell a story.
- Don't show the whole process, only the most important findings and decisions.
- Finish with a conclusion.
- Don't make lots of slides.

Good luck!